AUTOMATED CAR CATALOG SYSTEM FOR ENHANCED SHOWROOM MANAGEMENT

1. Document Overview

This document defines the functional specification of the Automated Car Catalog System for Enhanced Showroom Management developed using the ServiceNow platform.

The purpose of this document is to describe the functional behavior of the system, including:

- Car catalog management
- Customer request handling
- Role-based access control (RBAC)
- Group management
- Workflow automation
- Audit monitoring and reporting

The scope of this system is to automate showroom operations by reducing manual intervention in managing car models, approvals, and task assignments.

Intended Audience:

- Developers
- Testers
- Project Managers
- System Administrators
- Showroom Management Staff

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2. System Overview

The Automated Car Catalog System is a web-based showroom management solution designed to streamline car catalog operations and automate customer request processing.

The system enables showrooms to:

- Organize car models into structured categories
- Manage users and roles securely
- Automate approval workflows
- Track customer requests
- Monitor system activities

The platform uses Role-Based Access Control (RBAC) and workflow automation to ensure secure and efficient processing of showroom operations.

The primary objective of the system is to:

- Reduce manual effort
- Improve operational efficiency

- Enhance customer satisfaction
- Ensure transparency in approval processes

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3. Functional Modules

The system is divided into the following major functional modules:

3.1 Car Catalog Management Module

This module manages car models and categories.

Functions include:

- Add new car models
- Categorize vehicles (SUV, Sedan, Hatchback, Electric)
- Update pricing and availability
- Delete outdated models
- Search and filter vehicles

This module ensures centralized management of showroom inventory information.

3.2 User Management Module

This module handles the complete user lifecycle.

Functions:

- User registration
- Profile updates
- Activation and deactivation
- Password management
- User search

Types of users include:

- Admin
- Manager
- Sales Executive
- Customer

All user activities are recorded for monitoring.

3.3 Role and Group Management Module

This module enables administrators to define roles and manage permissions.

Functions include:

- Create roles
- Define access permissions
- Create user groups
- Assign users to groups
- Map roles to groups

Example:

- Admin → Full access
- Manager → Approval rights

- Sales Executive → Request handling
- Customer → View and submit requests

This structure simplifies access control and ensures security.

3.4 Access Control Module

The Access Control Module ensures secure authentication and authorization.

Functions:

- Validate login credentials
- Generate secure user sessions
- Check permissions before access
- Restrict unauthorized operations

This module prevents unauthorized access and protects sensitive data.

3.5 Workflow Management Module

This module automates the customer request and approval process.

Workflow Process:

1. Customer selects car and submits request
2. Request assigned to Sales Executive
3. Manager reviews request
4. Approval or rejection decision
5. Status update sent to customer

Features:

- Multi-level approvals
- Email notifications
- Approval history tracking
- Automatic task assignment

This reduces delays and ensures accountability.

3.6 Audit and Monitoring Module

This module records critical system activities such as:

- User logins
- Role modifications
- Approval actions
- Catalog updates

The system provides audit reports for compliance and security review.

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4. Functional Requirements Description

Each function of the system is defined with input, processing logic, and output.

4.1 Car Request Submission

Input:
Customer selects vehicle and submits request form.
Processing:

- Validate required fields
- Store request in database
- Trigger workflow
- Assign to Sales Executive

Output:

- Request ID generated
- Status set to “Pending Approval”
- Notification sent to assigned user

4.2 Approval Process

Input:
Manager reviews request.
Processing:

- Approve or reject decision
- Update status in database
- Record approval history

Output:

- Status updated (Approved/Rejected)
- Notification sent to customer

4.3 User Login

Input:
Username and password
Processing:

- Authentication validation
- Role verification
- Session creation

Output:

- Role-based dashboard displayed

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5. User Interface Requirements

The system interface must be:

- Responsive
- User-friendly
- Role-aware

Interface Components:

- Admin Dashboard
- Manager Approval Panel
- Sales Executive Task View

- Customer Service Portal
  The system must provide:
- Clear navigation
- Validation messages
- Status indicators
- Search functionality

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6. Security Requirements

The system must implement:

- Strong authentication mechanism
- Encrypted communication (HTTPS)
- Secure session handling
- Role-based authorization checks
- Activity logging
  Passwords must be securely stored.
  The design must prevent:
- Unauthorized access
- Privilege escalation
- Data manipulation

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7. Reporting Requirements

The system should generate:

- Car catalog reports
- Customer request reports
- Approval history reports
- User activity logs
- Audit trail reports
  Reports must be:
- Viewable on screen
- Downloadable for compliance review

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8. Assumptions and Dependencies

Assumptions:

- Stable internet connectivity
- Authorized personnel available
- ServiceNow platform properly configured
  Dependencies:
- Database server
- Hosting infrastructure
- Email notification service
- Web browser compatibility

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9. Acceptance Criteria

The system will be accepted when:

- All modules function correctly
- Workflows execute without errors
- Role-based access control works properly
- Reports generate accurately
- Security testing is successful
- User Acceptance Testing (UAT) is completed

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10. Problem Statement

Car showrooms and dealerships often face challenges in managing and streamlining their catalog of cars, including various models and categories.

The manual processes involved in handling customer requests, approvals, and task assignments can lead to inefficiencies, delays, and potential errors.

This can negatively impact customer satisfaction and operational efficiency.

The Automated Car Catalog System is designed to eliminate these challenges by introducing structured catalog management, automated workflows, and secure role-based access control.

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11. Conclusion

The Car Catalog System project for a car showroom in ServiceNow successfully addresses the need for a streamlined, automated approach to managing car catalog items, customer requests, and approval workflows.

By organizing car models into easily navigable categories and leveraging ServiceNow's powerful automation features, the system:

- Enhances operational efficiency
- Reduces manual intervention
- Improves transparency
- Increases customer satisfaction

The solution provides a scalable, secure, and efficient showroom management system that meets modern operational requirements.